Before class: chains, names and double bonds

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

Saturated and unsaturated

Alkanes have only single C–C bonds: every carbon holds as many hydrogens as it can, so they are **saturated**. **Alkenes** (C=C) and **alkynes** (C≡C) have fewer hydrogens: they are **unsaturated**. **Benzene** is a special ring of six carbons called an **aromatic** compound.

Names come from counting carbons

The first four stems are worth learning by heart: **meth**- (1 C), **eth**- (2), **prop**- (3), **but**- (4). Then pent-, hex-, hept-, oct-, non-, dec- (5 to 10). The ending tells the family: **-ane** (alkane), **-ene** (alkene), **-yne** (alkyne). So propane, propene and propyne all have three carbons.

Why this matters in pharmacy

Petroleum jelly and liquid paraffin are alkanes. IV bags and syringes are made from alkene polymers. Many drugs contain a benzene ring. And the gas cylinder in your kitchen holds propane and butane.

2 · Readiness check

Ten quick questions: Chapter 1 recap and a preview.

Q1 Recap

How many bonds does each carbon form in a neutral organic molecule?

A. 2
B. 3
C. 4
D. 6

Q2 Recap

How many hydrogen atoms are on a carbon at the end of a zig-zag line (no other atom written)?

Q3 Recap

What is the shape around a carbon with a C=C double bond?

A. tetrahedral, 109.5°
B. trigonal planar, 120°
C. linear, 180°
D. bent, 104.5°

Q4 Recap

Which intermolecular force holds alkane molecules together?

A. hydrogen bonds
B. ionic bonds
C. dispersion (London) forces
D. covalent bonds

Q5 CH2-LO1

Match each stem to its number of carbons.

Options: 1 · 2 · 3 · 4 · 5 · 6

meth- _____

eth- _____

prop- _____

but- _____

pent- _____

Q6 CH2-LO1

Alkanes have the general formula C_nH_{2n+2}. How many H atoms are in an alkane with 6 carbons?

_____

Q7 CH2-LO4

Which name belongs to an alkene?

A. propane

B. propene

C. propyne

D. propanol

Q8 CH2-LO2

Oil floats on water because alkanes are nonpolar and less dense than water.

True / False

Q9 CH2-LO3

What are the products of the complete combustion of an alkane?

A. CO and H_2

B. CO_2 and H_2O

C. C and H_2O

D. CH_4 and O_2

Q10 CH2-LO5

Margarine is made by adding hydrogen to vegetable oil. Guess what happens to the C=C bonds in the oil.

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Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.3, 12.6–12.10 and exercises 12.E (Sharpe Elles, from Ball, Hill & Scott), *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 13 and *Fundamentals of General Organic and Biological Chemistry* 13.1–13.9, CC BY-NC-SA 3.0. Reaction mechanisms, Markovnikov items and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Saturated and Unsaturated Hydrocarbons

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 Recap

How many bonds does each carbon form in a neutral organic molecule?

A. 2

B. 3

C. 4

D. 6

Carbon is tetravalent.

Q2 Recap

How many hydrogen atoms are on a carbon at the end of a zig-zag line (no other atom written)?

3

One bond to carbon, so 3 H: a CH_3 group.

Q3 Recap

What is the shape around a carbon with a C=C double bond?

A. tetrahedral, 109.5°

B. trigonal planar, 120°

C. linear, 180°

D. bent, 104.5°

Three electron groups.

Q4 Recap

Which intermolecular force holds alkane molecules together?

A. hydrogen bonds

B. ionic bonds

C. dispersion (London) forces

D. covalent bonds

Alkanes are nonpolar, so only dispersion forces act between them.

Q5 CH2-LO1

Match each stem to its number of carbons.

meth- → 1; eth- → 2; prop- → 3; but- → 4; pent- → 5

Learn the first ten stems: they are used for every family.

Q6 CH2-LO1

Alkanes have the general formula C_nH_{2n+2}. How many H atoms are in an alkane with 6 carbons?

14

2(6) + 2 = 14: hexane, C_6H_{14}.

Q7 CH2-LO4

Which name belongs to an alkene?

A. propane

B. propene

C. propyne

D. propanol

-ene means a C=C double bond.

Q8 CH2-LO2

Oil floats on water because alkanes are nonpolar and less dense than water.

True / False → **True**

Most liquid alkanes have densities of about 0.6–0.7 g/mL.

Q9 CH2-LO3

What are the products of the complete combustion of an alkane?

A. CO and H_2

B. CO_2 and H_2O

C. C and H_2O

D. CH_4 and O_2

With plenty of oxygen, alkanes burn to carbon dioxide and water.

Q10 CH2-LO5

Margarine is made by adding hydrogen to vegetable oil. Guess what happens to the C=C bonds in the oil.

Hydrogen adds across the C=C double bonds, turning them into C–C single bonds (hydrogenation). The oil becomes more saturated and more solid. You will meet this addition reaction in 2.2.

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